1 Lyapunov's stability theorem and Barbalat's lemma

Definition 1. *Consider the dynamical system $\dot{x} = f(x,u)$ with trajectory $x(t) = \varphi(t,x,u)$. An equilibrium (x_e, u_e) is* ***stable*** *if a sufficiently small perturbation of the initial state x_0 keeps the trajectory within any prescribed distance of x_e.*

$$\forall \varepsilon > 0 \quad \exists \delta_\varepsilon > 0 \quad / \quad \|x_0 - x_e\| < \delta_\varepsilon \quad \Rightarrow \quad \|\varphi(t, x_0, u_e) - x_e\| < \varepsilon \quad \forall t \geqslant 0$$

Definition 2. *Consider the dynamical system $\dot{x} = f(x,u)$ with trajectory $x(t) = \varphi(t,x,u)$. An equilibrium (x_e, u_e) is* ***exponentially stable*** *if, after a sufficiently small perturbation of the initial state x_0 , the trajectories converge exponentially to x_0.*

$$\forall \varepsilon > 0 \quad \exists \delta_\varepsilon > 0 \quad / \quad \|x_0 - x_e\| < \delta_\varepsilon \quad \Rightarrow \quad \|\varphi(t, x_0, u_e) - x_e\| < \varepsilon \quad \forall t \geqslant 0$$

$$\exists K > 0, \lambda > 0, \delta_e > 0 \quad / \quad \|x_0 - x_e\| < \delta_e \quad \Rightarrow \quad \|\varphi(t, x_0, u_e) - x_e\| < K\, e^{-\lambda t} \quad \forall t \geqslant 0$$

Theorem 3. ***(Lyapunov's stability theorem)***

Consider the autonomous system $\dot{x} = f(x)$ with equilibrium $x_e = 0$. If $\exists\ V \in C^1$, $V>0$, such that $\dot{V} \leq 0$, then the equilibrium $x_e = 0$ is stable.

Theorem 4. ***(Barbalat's lemma)***

If a function $r(t) \in L_2 \cap L_\infty$ has a bounded derivative ($\dot{r}(t) \in L_\infty$), it converges asymptotically to zero.

2 Linear parameterisation

Consider a SISO system

$$\begin{aligned} \dot{x} &= A\,x + B\,u & x \in \mathbb{R}^n, u \in \mathbb{R} \\ y &= C\,x & y \in \mathbb{R} \end{aligned}$$

with unknown matrices A, B and C. A **linear parameterisation** is a relation of the form

$$z(t) = \theta^T \phi(t)$$

where

- $z(t)$ is a known, measurable signal
- θ contains all the unknown constant parameters
- $\phi(t)$ is called the **regressor vector** and contains only known or measurable signals

2.1 Parameterisation in the Laplace domain

Assuming zero initial conditions $x(0) = 0$, we obtain

$$\begin{aligned} s\,x(s) &= A\,x(s) + B\,u(s) \\ y(s) &= C\,x(s) \end{aligned}$$

$$\begin{aligned} x(s) &= (s\,I - A)^{-1} B\,u(s) \\ y(s) &= C\,(s\,I - A)^{-1} B\,u(s) \\ &= G(s)\,u(s) \end{aligned}$$

where $G(s)$ is the **input-output transfer matrix**.

Since

$$(s\,I - A)^{-1} = \frac{1}{\det(s\,I - A)} (s\,I - A)^*$$

we can make the following observations.

- Since A has n eigenvalues λ_i and $\det(\mathrm{sI} - A) = \prod_{i=1}^{n} (s - \lambda_i)$, the polynomial $\det(s\,I - A)$ has degree n. The denominator of the input-output transfer matrix can therefore be written as

$$\mathrm{den}(G(S)) = s^n + a_{n-1} \cdot s^{n-1} + \cdots + a_1 s + a_0$$

- The entry a^*_{ij} of the adjugate matrix $(s\,I - A)^*$ is defined as

$$a^*_{\mathrm{ij}} = (-1)^{i+j} \det(A_{\mathrm{ji}})$$

  The numerator of the input-output transfer matrix therefore has degree at most $n - 1$ and can be written as

$$\mathrm{num}(G(S)) = b_{n-1} \cdot s^{n-1} + b_{n-2} \cdot s^{n-2} + \cdots + b_1 s + b_0$$

It follows that

$$\begin{aligned}
y(s) &= \frac{b_{n-1}\cdot s^{n-1}+b_{n-2}\cdot s^{n-2}+\cdots+b_1 s+b_0}{s^n+a_{n-1}\cdot s^{n-1}+\cdots+a_1 s+a_0}\,u(s) \\
(s^n+a_{n-1}\cdot s^{n-1}+\cdots+a_1 s+a_0)y(s) &= (b_{n-1}\cdot s^{n-1}+b_{n-2}\cdot s^{n-2}+\cdots+b_1 s+b_0)u(s) \\
s^n y(s)+a_{n-1}\cdot s^{n-1}y(s)+\cdots+a_1\, s\, y(s)+a_0 y(s) &= b_{n-1}s^{n-1}\,u(s)+\cdots+b_1 s\,u(s)+b_0 u(s) \\
s^n y(s) &= b_{n-1}\,s^{n-1}\,u(s)+\cdots+b_1 s\,u(s)+b_0 u(s) \\
&\quad -a_{n-1}\,s^{n-1}y(s)-\cdots-a_1\,s\,y(s)-a_0 y(s)
\end{aligned}$$

Define

$$\theta^T \;=\; (\; b_{n-1} \;\; \ldots \;\; b_1 \;\; b_0 \;\; a_{n-1} \;\; \ldots \;\; a_1 \;\; a_0 \;)$$

Then

$$\begin{aligned}
s^n y(s) &= (\; b_{n-1} \;\; \ldots \;\; b_1 \;\; b_0 \;\; a_{n-1} \;\; \ldots \;\; a_1 \;\; a_0 \;)\begin{pmatrix} s^{n-1}u(s) \\ \vdots \\ s\,u(s) \\ u(s) \\ -s^{n-1}y(s) \\ \vdots \\ -s\,y(s) \\ -y(s) \end{pmatrix} \\
&= (\; b_{n-1} \;\; \ldots \;\; b_1 \;\; b_0 \;\; a_{n-1} \;\; \ldots \;\; a_1 \;\; a_0 \;)\begin{pmatrix} \alpha_{n-1}(s)u(s) \\ -\alpha_{n-1}(s)y(s) \end{pmatrix}
\end{aligned}$$

where

$$\alpha_{n-1}(s) \;=\; \begin{pmatrix} s^{n-1} \\ \vdots \\ s \\ 1 \end{pmatrix}$$

This expression is already linear in the unknown parameters, but it contains terms of the form $s^i u(s)$ and $s^i y(s)$. In the time domain, these terms correspond to the i-th derivatives of $y(t)$ and $u(t)$. Since these derivatives can be difficult to measure or compute reliably, we divide both sides by a Hurwitz polynomial of degree n

$$\Lambda(s) \;=\; s^n+\lambda_{n-1}s^{n-1}+\cdots+\lambda_1 s+\lambda_0$$

to obtain

$$\begin{aligned}
\frac{s^n y(s)}{\Lambda(s)} &= \frac{1}{\Lambda(s)}(\; b_{n-1} \;\; \ldots \;\; b_1 \;\; b_0 \;\; a_{n-1} \;\; \ldots \;\; a_1 \;\; a_0 \;)\begin{pmatrix} \alpha_{n-1}(s)u(s) \\ -\alpha_{n-1}(s)y(s) \end{pmatrix} \\
&= (\; b_{n-1} \;\; \ldots \;\; b_1 \;\; b_0 \;\; a_{n-1} \;\; \ldots \;\; a_1 \;\; a_0 \;)\begin{pmatrix} \frac{\alpha_{n-1}(s)}{\Lambda(s)}u(s) \\ -\frac{\alpha_{n-1}(s)}{\Lambda(s)}y(s) \end{pmatrix} \\
&= (\; b_{n-1} \;\; \ldots \;\; b_1 \;\; b_0 \;\; a_{n-1} \;\; \ldots \;\; a_1 \;\; a_0 \;)\begin{pmatrix} \xi(s) \\ \eta(s) \end{pmatrix}
\end{aligned}$$

2.2 Realisation of the regressor vector $\phi(t)$

To realise the filtered signals

$$\xi(s) \;=\; \frac{\alpha_{n-1}(s)}{\Lambda(s)}u(s)$$

use the state-space realisation

$$\dot{\xi}(t) \;=\; A_u\,\xi(t) + b_u\,u(t)$$

where A_u is in observable canonical form and is given by

$$A_u \;=\; \begin{pmatrix} -\lambda_{n-1} & -\lambda_{n-2} & \dots & -\lambda_1 & -\lambda_0 \\ 1 & 0 & & & \\ 0 & 1 & \ddots & & \\ & & \ddots & 0 & \\ & & & 1 & 0 \end{pmatrix}$$

$$b_u \;=\; \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Indeed,

$$\begin{aligned} \xi(s) &= (s\,I - A_u)^{-1} b_u \xi(s) \\ &= \frac{1}{\det(s\,I - A_u)}(s\,I - A_u)^* \cdot e_1 \cdot u(s) \end{aligned}$$

where

$$\begin{aligned} (s\,I - A_u)^* &= \begin{pmatrix} s+\lambda_{n-1} & \lambda_{n-2} & \dots & \lambda_1 & \lambda_0 \\ -1 & s & & & \\ 0 & -1 & \ddots & & \\ & & \ddots & s & \\ & & & -1 & s \end{pmatrix}^* \\ &= \begin{pmatrix} s^{n-1} & * \\ -(-s^{n-2}) & * \\ \vdots & \vdots \\ 1 & * \end{pmatrix} \end{aligned}$$

and

$$\begin{aligned} \det(s\,I - A_u) &= (s+\lambda_{n-1})\,s^{n-1} - \lambda_{n-2}(-s^{n-2}) + \dots + \lambda_0 \\ &= s^n + \lambda_{n-1}s^{n-1} + \lambda_{n-2}s^{n-2} + \dots + \lambda_0 \\ &= \Lambda(s) \end{aligned}$$

It follows that

$$\begin{aligned} \xi(s) &= \frac{1}{\det(s\,I - A_u)}(s\,I - A_u)^* e_1 u(s) \\ &= \frac{1}{\Lambda(s)} \begin{pmatrix} s^{n-1} \\ -(-s^{n-2}) \\ \vdots \\ 1 \end{pmatrix} u(s) \\ &= \frac{\alpha_{n-1}(s)}{\Lambda(s)}u(s) \end{aligned}$$

The same reasoning gives a realisation for the dynamics of $\eta(s) = -\frac{\alpha_{n-1}(s)}{\Lambda(s)} y(s)$

$$\dot{\eta}(t) \;=\; A_y \xi(t) + b_y y(t)$$

by setting

$$\begin{aligned} A_y &= A_u \\ b_y &= -b_u = \begin{pmatrix} -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \end{aligned}$$

2.3 Realisation of the known signal $z(t)$

Finally, to realise

$$z(t) \;=\; L^{-1}\left[\frac{s^n}{\Lambda(s)} y(s)\right]$$

recall that

$$\Lambda(s) \;=\; s^n + \lambda_{n-1} s^{n-1} + \cdots + \lambda_1 s + \lambda_0$$

or equivalently,

$$s^n \;=\; \Lambda(s) - \lambda_{n-1} s^{n-1} - \cdots - \lambda_1 s - \lambda_0$$

It follows that

$$\begin{aligned} z(s) &= \frac{s^n}{\Lambda(s)} y(s) \\ &= \frac{\Lambda(s) - \lambda_{n-1} s^{n-1} - \cdots - \lambda_1 s - \lambda_0}{\Lambda(s)} \cdot y(s) \\ &= \left(1 - \frac{\lambda_{n-1} s^{n-1} + \cdots + \lambda_1 s + \lambda_0}{\Lambda(s)}\right) \cdot y(s) \\ &= y(s) - \lambda_{n-1} \frac{s^{n-1}}{\Lambda(s)} y(s) - \cdots - \lambda_1 \frac{s}{\Lambda(s)} y(s) - \lambda_0 \frac{1}{\Lambda(s)} y(s) \\ &= y(s) + \begin{pmatrix} \lambda_{n-1} & \dots & \lambda_1 & \lambda_0 \end{pmatrix} \begin{pmatrix} -\frac{s^{n-1}}{\Lambda(s)} y(s) \\ \vdots \\ -\frac{s}{\Lambda(s)} y(s) \\ -\frac{1}{\Lambda(s)} y(s) \end{pmatrix} \\ &= y(s) - \begin{pmatrix} \lambda_{n-1} & \dots & \lambda_1 & \lambda_0 \end{pmatrix} \left(-\frac{\alpha_{n-1}(s)}{\Lambda(s)} y(s)\right) \\ &= y(s) + \begin{pmatrix} \lambda_{n-1} & \dots & \lambda_1 & \lambda_0 \end{pmatrix} \eta(s) \end{aligned}$$

Therefore,

$$z(t) \;=\; y(t) + \begin{pmatrix} \lambda_{n-1} & \dots & \lambda_1 & \lambda_0 \end{pmatrix} \eta(t)$$

3 Gradient identifier

Consider a linear SISO system

$$\begin{aligned} \dot{x} &= A\,x + B\,u \\ y &= C\,x \end{aligned}$$

with the linear parameterisation

$$z(t) = \theta^T \phi(t)$$

The **gradient method** provides an estimate of the parameter vector θ whose error converges asymptotically to zero.

Given the estimate of the linear parameterisation

$$\hat{z}(t) = \hat{\theta}^T \phi(t)$$

the parameterisation estimation error is

$$\begin{aligned} e(t) &= \hat{z}(t) - z(t) \\ &= \hat{\theta}^T \phi(t) - z(t) \\ &= \tilde{\theta}^T \phi(t) \end{aligned}$$

where

$$\tilde{\theta} = \hat{\theta} - \theta$$

To minimise the error, the gradient method introduces the cost function

$$\begin{aligned} J(\hat{\theta}) &= \frac{1}{2} e^2(t) \\ &= \frac{1}{2} (\hat{\theta}^T \phi(t) - z(t))^2 \end{aligned}$$

and makes the dynamics of the parameter estimate $\hat{\theta}$ follow the negative gradient direction.

$$\dot{\hat{\theta}} = -\Gamma \cdot \nabla J(\hat{\theta})$$

Here, $\Gamma = \Gamma^T > 0$ is the **adaptation gain matrix** and determines how quickly the parameter estimates are updated. Smaller values of Γ lead to slower updates, while larger values of Γ lead to faster updates.

The gradient of J, expressed as a column vector, is

$$\begin{aligned} \nabla J(\hat{\theta}) &= \nabla \left[\frac{1}{2} (\hat{\theta}^T \phi(t) - z(t))^2 \right] \\ &= \frac{1}{2} 2 (\hat{\theta}^T \phi(t) - z(t))\, \phi(t) \\ &= (\hat{\theta}^T \phi(t) - z(t))\, \phi(t) \\ &= e(t)\, \phi(t) \end{aligned}$$

so the adaptation law is

$$\dot{\hat{\theta}} = -\Gamma \cdot e(t) \cdot \phi(t)$$

Definition 5. *A signal* $u(t)\colon \mathbb{R}^{\geq 0} \to \mathbb{R}^m$ *is* ***persistently exciting*** *if there exist three constants* $t_0 > 0, T > 0, \alpha > 0$ *such that*

$$\frac{1}{T}\int_t^{t+T} u(\tau)\cdot u^T(\tau)d\tau \geq \alpha I > 0 \qquad \forall t \geq t_0$$

Theorem 6. *Consider the SISO system*

$$\begin{aligned} \dot{x}(t) &= A\,x(t) + b\,u(t) \\ y(t) &= C\,x(t) \end{aligned}$$

and the parameterisation

$$z(t) = \theta^T \phi(t)$$

If the regressor vector $\phi(t)$ *is bounded and the gradient method is used to estimate the parameters* θ *according to*

$$\dot{\hat{\theta}} = -\Gamma \cdot e(t) \cdot \phi(t)$$

then the parameterisation estimation error $e(t)$ *converges asymptotically to zero.*

$$e(t) \to 0$$

Moreover, if ϕ *is persistently exciting, the parameter estimate* $\hat{\theta}(t)$ *converges exponentially to* θ*.*

Definition 7. *We prove only that* $e(t) \to 0$*.*

The regressor vector is

$$\phi(t) = \begin{pmatrix} \xi(t) \\ \eta(t) \end{pmatrix}$$

where

$$\dot{\xi}(t) = A_u \xi(t) + u(t)$$

and

$$\dot{\eta}(t) = A_u \eta(t) - y(t)$$

Since $\phi \in L_\infty$ *by assumption, we have* $\xi(t), \eta(t), u(t), y(t) \in L_\infty$*. It follows that* $\dot{\phi}(t) = \begin{pmatrix} \dot{\xi}(t) \\ \dot{\eta}(t) \end{pmatrix} \in L_\infty$*.*

Since $\Gamma = \Gamma^T > 0$*, Sylvester's criterion implies that* Γ *is non-singular* $\det \Gamma \neq 0$*. Define*

$$V(\tilde{\theta}) = \frac{1}{2}\tilde{\theta}^T \Gamma^{-1} \tilde{\theta}$$

where

$$\tilde{\theta} = \hat{\theta} - \theta$$

Then

$$\begin{aligned}
\dot{V}(\tilde{\theta}) &= \frac{1}{2}\dot{\tilde{\theta}}^T\Gamma^{-1}\tilde{\theta} + \frac{1}{2}\tilde{\theta}^T\Gamma^{-1}\dot{\tilde{\theta}} \\
&= \left(\frac{1}{2}\dot{\tilde{\theta}}^T\Gamma^{-1}\tilde{\theta}\right)^T + \frac{1}{2}\tilde{\theta}^T\Gamma^{-1}\dot{\tilde{\theta}} \\
&= \tilde{\theta}^T\Gamma^{-1}\dot{\tilde{\theta}} \\
&= \tilde{\theta}^T\Gamma^{-1}(-\Gamma\cdot e(t)\cdot\phi(t)) \\
&= -\tilde{\theta}^T\Gamma\not{^{-1}}\not{\Gamma}\cdot e(t)\cdot\phi(t) \\
&= -\tilde{\theta}^T\cdot e(t)\cdot\phi(t) \\
&= -\tilde{\theta}^T\cdot\phi(t)\cdot e(t) \\
&= -e^2(t) \\
&\leq 0
\end{aligned}$$

Lyapunov's stability theorem therefore gives $\tilde{\theta}\in L_\infty$.

Since $e=\tilde{\theta}^T\cdot\phi(t)$, *it also follows that* $e\in L_\infty$.

Since

$$\begin{aligned}
\|e\|_2^2 &= \int_0^\infty e^2(t)\mathrm{dt} \\
&= \int_0^\infty -\dot{V}(\tilde{\theta})\mathrm{dt} \\
&= -V(\tilde{\theta}(\infty)) + V(\tilde{\theta}(0))
\end{aligned}$$

and since $V(\tilde{\theta}(t))\geq 0 \quad \forall t\geq 0$ *and* $V(\tilde{\theta}(0))<\infty$, *we obtain*

$$\begin{aligned}
\|e\|_2^2 &= -V(\tilde{\theta}(\infty)) + V(\tilde{\theta}(0)) \\
&\leq V(\tilde{\theta}(0)) \\
&< \infty
\end{aligned}$$

or equivalently $e\in L_2$.

Finally, since

$$\begin{aligned}
\dot{e} &= \dot{\tilde{\theta}}^T\phi + \tilde{\theta}^T\dot{\phi} \\
&= (-\Gamma\phi e)^T\phi + \tilde{\theta}^T\dot{\phi}
\end{aligned}$$

and $\phi, e, \theta, \dot{\phi}\in L_\infty$, *we obtain* $\dot{e}\in L_\infty$.

By Barbalat's lemma,

$$e(t) \;\rightarrow\; 0$$

This proves the result.

Remark 8. Recall that the error $e=\tilde{\theta}^T\phi(t)$ is a scalar and

$$\begin{aligned}
\dot{\tilde{\theta}} &= -\Gamma\cdot e(t)\cdot\phi(t) \\
&= -\Gamma\cdot\phi(t)\cdot e(t) \\
&= -\Gamma\cdot\phi(t)\cdot\phi^T(t)\cdot\tilde{\theta}
\end{aligned}$$

The matrix $\phi(t)\phi^T(t)$ therefore appears in the adaptation law, which means that the *errors* for the different parameters are *coupled*.

For example, for a linear parameterisation with two parameters,

$$\theta = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \Gamma = \begin{pmatrix} \gamma_1 & 0 \\ 0 & \gamma_2 \end{pmatrix} \qquad \phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

we obtain

$$\begin{aligned}
\dot{\tilde{\theta}} &= \dot{\hat{\theta}} \\
&= -\Gamma \cdot \phi(t) \cdot \phi^T(t) \cdot \tilde{\theta} \\
&= -\begin{pmatrix} \gamma_1 & 0 \\ 0 & \gamma_2 \end{pmatrix}\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}\begin{pmatrix} \phi_1 & \phi_2 \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix} \\
&= -\begin{pmatrix} \gamma_1 & 0 \\ 0 & \gamma_2 \end{pmatrix}\begin{pmatrix} \phi_1^2 & \phi_1\phi_2 \\ \phi_1\phi_2 & \phi_2^2 \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix} \\
&= -\begin{pmatrix} \gamma_1\phi_1^2 & \gamma_1\phi_1\phi_2 \\ \gamma_2\phi_1\phi_2 & \gamma_2\phi_2^2 \end{pmatrix}\begin{pmatrix} \tilde{a} \\ \tilde{b} \end{pmatrix} \\
&= -\begin{pmatrix} \gamma_1\phi_1^2\tilde{a} + \gamma_1\phi_1\phi_2\tilde{b} \\ \gamma_2\phi_1\phi_2\tilde{a} + \gamma_2\phi_2^2\tilde{b} \end{pmatrix} \\
\begin{pmatrix} \dot{\tilde{a}} \\ \dot{\tilde{b}} \end{pmatrix} &= \begin{pmatrix} \gamma_1(-\phi_1^2\hat{a} - \phi_1\phi_2\hat{b} + \phi_1) \\ \gamma_2(-\phi_1\phi_2\hat{a} - \phi_2^2\hat{b} + \phi_2) \end{pmatrix}
\end{aligned}$$

The dynamics of $\tilde{a}$ therefore depend on $\tilde{b}$, and vice versa.

4 DREM

In the gradient method, the parameter estimation errors are coupled, so each parameter estimate depends on the others. **DREM** (**Dynamic Regressor Extension and Mixing**) is an adaptive estimation technique that transforms the multivariable parameterisation

$$z(t) \;=\; \theta^T \phi(t) \qquad \theta \in \mathbb{R}^q$$

into independent scalar parameterisations. The procedure is as follows.

1. Apply a stable filter $\mathcal{H}(s) \in \mathbb{R}^q$ of the form

   $$\mathcal{H}_i(s) = \frac{b_i}{s+\tau_i} \qquad \vee \qquad \mathcal{H}_i(s) = e^{-\tau_i s} \qquad \forall i = 1 \ldots q$$

   with $\tau_i > 0, b_i \neq 0$ and $\tau_i \neq \tau_j \, \forall i \neq j$, obtaining the signals

   $$Z(s) \;=\; \mathcal{H}(s) z(s) \quad \in \quad \mathbb{R}^q$$

   and

   $$\Phi(s) \;=\; \mathcal{H}(s) \, \phi^T(s) \quad \in \quad \mathbb{R}^q \times \mathbb{R}^{1 \times q} = \mathbb{R}^{q \times q}$$

2. In the time domain, determine the dynamics of the components of $Z(t)$ and $\Phi(t)$ in terms of $z(t)$ and $\phi(t)$.

3. For a square matrix $\Phi(t) \in \mathbb{R}^{q \times q}$, recall that

   $$\Phi^*(t) \, \Phi(t) \;=\; \det\{\Phi(t)\} I$$

   Multiply both sides of $Z(t) = \Phi(t)\theta$ by $\Phi^*(t)$ to obtain

   $$\begin{aligned} \Phi^*(t) Z(t) \;&=\; \Phi^*(t) \Phi(t) \theta \\ &=\; \det(\Phi(t)) \, \theta \end{aligned}$$

4. Set $\tilde{Z}_i(t) = [\Phi^*(t) Z(t)]_i$ to obtain q decoupled inequalities.

   $$\bar{Z}_i(t) \;=\; \det(\Phi(t)) \theta_i$$

Theorem 9. *Consider a system with a linear parameterisation decoupled using DREM*

$$\bar{Z}_i(t) \;=\; \det(\Phi(t)) \theta_i \quad i = 1 \ldots p$$

If the gradient method is used to estimate the parameters according to

$$\dot{\hat{\theta}}_i \;=\; -\gamma_i \cdot e_i(t) \cdot \det(\Phi(t))$$

with $\gamma_i > 0$, then

$$|\tilde{\theta}_i(t)| \;\leq\; |\tilde{\theta}_i(0)| \qquad \forall t \geq 0$$

Moreover,

$$\tilde{\theta} \to 0 \qquad \Leftrightarrow \qquad \det \Phi \notin L_2$$

Proof.

The estimation error for each parameter is

$$\begin{aligned} e_i(t) &= \hat{\bar{Z}}_i(t) - \bar{Z}_i(t) \\ &= \det(\Phi(t)) \cdot \hat{\theta}_i - \det(\Phi(t)) \cdot \theta_i \\ &= \det(\Phi(t)) \cdot \tilde{\theta}_i(t) \end{aligned}$$

Applying the gradient method to estimate the parameters then gives

$$\begin{aligned} \dot{\tilde{\theta}}_i &= -\gamma_i \cdot e_i(t) \cdot \det(\Phi(t)) \\ &= -\gamma_i \cdot \tilde{\theta}_i(t) \cdot \det^2(\Phi(t)) \end{aligned}$$

or equivalently,

$$\tilde{\theta}_i(t) = \theta_i(0)\, e^{-\gamma_i \int_0^t \det^2(\Phi(t))\mathrm{dt}}$$

Since $e^{-\gamma_i \int_0^t \det^2(\Phi(t))\mathrm{dt}}$ is a decreasing function,

$$|\tilde{\theta}_i(t)| \leq |\tilde{\theta}_i(0)| \qquad \forall t \geq 0$$

Moreover,

$$\begin{aligned} \tilde{\theta}_i(t) &\to 0 \\ \Updownarrow & \\ e^{-\gamma_i \int_0^t \det^2(\Phi(t))\mathrm{dt}} &\to 0 \\ \Updownarrow & \\ -\gamma_i \int_0^t \det^2(\Phi(t))\mathrm{dt} &\to -\infty \\ \Updownarrow & \\ \int_0^\infty \det^2(\Phi(t))\mathrm{dt} &= \infty \\ \Updownarrow & \\ \det^2(\Phi(t)) &\notin L_2 \end{aligned}$$

This proves the result. □

5 Assignment

Consider the system

$$\dot{x} = -ax + bu$$

with unknown parameters $a > 0, b \neq 0$.

Find a parameterisation of the system and design a DREM parameter estimator and a gradient identifier.

Simulate the DREM and gradient estimators assuming

$$a = 0.4, \qquad b = 0.4,$$

and use the following filters for the DREM estimator.

$$H_1(s) = \frac{1}{s+1}, \qquad H_2(s) = \frac{2}{s+2}.$$

Consider the two inputs

$$u(t) = 10$$

and

$$u(t) = 10 \sin\left(\frac{5t}{2}\right)$$

Compare the estimators in terms of

- transient response
- convergence rate
- overall performance.

Also plot the phase portraits of the parameter estimation errors and discuss the results.

6 Estimator design

6.1 Parameterisation

For $\dot{x} = -a\,x + b\,u$, we obtain

$$s\,x(s) \;=\; -a\,x(s) + b\,u(s)$$

Dividing both sides by the Hurwitz polynomial $\Lambda(s) = s+1$ gives

$$\begin{aligned}\frac{s\,x(s)}{s+1} &= \frac{-a\,x(s)+b\,u(s)}{s+1}\\ &= \begin{pmatrix} b & a \end{pmatrix}\begin{pmatrix} \frac{1}{s+1}u(s) \\ -\frac{1}{s+1}y(s) \end{pmatrix}\end{aligned}$$

First, realise the dynamics of the regressor vector $\begin{pmatrix} \xi \\ \eta \end{pmatrix}$. We obtain

$$\begin{aligned}\xi(s) &= \frac{1}{s+1}u(s)\\ (s+1)\xi(s) &= u(s)\\ s\cdot\xi(s) &= -\xi(s)+u(s)\\ &\downarrow\\ \dot{\xi}(t) &= -\xi(t)+u(t)\end{aligned}$$

and

$$\begin{aligned}\eta(s) &= -\frac{1}{s+1}x(s)\\ (s+1)\eta(s) &= -x(s)\\ s\cdot\eta(s) &= -\eta(s)-x(s)\\ &\downarrow\\ \dot{\eta}(t) &= -\eta(t)-x(t)\end{aligned}$$

To realise the output $z(t) = L^{-1}\Big[\frac{s}{s+1}x(s)\Big]$, note that

$$\begin{aligned}z(s) &= \frac{s}{s+1}x(s)\\ &= \frac{s+1-1}{s+1}x(s)\\ &= \left(1-\frac{1}{s+1}\right)x(s)\\ &= x(s)-\frac{1}{s+1}x(s)\\ &= x(s)+\eta(s)\\ &\downarrow\\ z(t) &= x(t)+\eta(t)\end{aligned}$$

In summary,

$$\begin{aligned}\theta &= \begin{pmatrix} b & a \end{pmatrix}\\ \phi(t) &= \begin{pmatrix} \xi(t) \\ \eta(t) \end{pmatrix}\\ \dot{\xi}(t) &= -\xi(t)+u(t)\\ \dot{\eta}(t) &= -\eta(t)-x(t)\\ z(t) &= x(t)+\eta(t)\end{aligned}$$

6.2 Gradient method

For the gradient method, set

$$\begin{aligned} e(t) &= \hat{\theta}^T\phi(t) - z(t) \\ \dot{\hat{\theta}} &= -\Gamma\, e(t)\, \phi(t) \end{aligned}$$

6.3 DREM

- First, compute the dynamics of

$$Z(s) = \mathcal{H}(s)z(s) = \begin{pmatrix} \frac{1}{s+1} \\ \frac{2}{s+2} \end{pmatrix} z(s)$$

  We obtain

$$\begin{aligned} Z_1(s) = \frac{1}{s+1}z(s) \qquad &\Rightarrow \qquad \dot{Z}_1(t) = -Z_1(t) + z(t) \\ Z_2(s) = \frac{2}{s+2}z(s) \qquad &\Rightarrow \qquad \dot{Z}_2(t) = -2Z_2(t) + 2z(t) \end{aligned}$$

- Next, compute the dynamics of $\Phi(s) = \mathcal{H}(s)\phi^T(s)$. We obtain

$$\begin{aligned} \Phi(s) &= \begin{pmatrix} \frac{1}{s+1} \\ \frac{2}{s+2} \end{pmatrix} \begin{pmatrix} \xi(s) & \eta(s) \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{s+1}\xi(s) & \frac{1}{s+1}\eta(s) \\ \frac{2}{s+2}\xi(s) & \frac{2}{s+2}\eta(s) \end{pmatrix} \end{aligned}$$

  It follows that

$$\begin{aligned} \Phi_{11}(s) = \frac{1}{s+1}\xi(s) \qquad &\Rightarrow \qquad \dot{\Phi}_{11}(t) = -\Phi_{11}(t) + \xi(t) \\ \Phi_{12}(s) = \frac{1}{s+1}\eta(s) \qquad &\Rightarrow \qquad \dot{\Phi}_{12}(t) = -\Phi_{12}(t) + \eta(t) \\ \Phi_{21}(s) = \frac{2}{s+2}\xi(s) \qquad &\Rightarrow \qquad \dot{\Phi}_{21}(t) = -2\,\Phi_{21}(t) + 2\,\xi(t) \\ \Phi_{22}(s) = \frac{2}{s+2}\eta(s) \qquad &\Rightarrow \qquad \dot{\Phi}_{22}(t) = -2\,\Phi_{22}(t) + 2\,\eta(t) \end{aligned}$$

  or equivalently,

$$\begin{aligned} \dot{\Phi}(t) &= \begin{pmatrix} -\Phi_{11}(t) + \xi(t) & -\Phi_{12}(t) + \eta(t) \\ -2\,\Phi_{21}(t) + 2\,\xi(t) & -2\,\Phi_{22}(t) + 2\,\eta(t) \end{pmatrix} \\ &= \begin{pmatrix} -\Phi_{11}(t) & -\Phi_{12}(t) \\ -2\,\Phi_{21}(t) & -2\,\Phi_{22}(t) \end{pmatrix} + \begin{pmatrix} \xi(t) & \eta(t) \\ 2\xi(t) & 2\,\eta(t) \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} \Phi_{11}(t) & \Phi_{12}(t) \\ \Phi_{21}(t) & \Phi_{22}(t) \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} \xi(t) & \eta(t) \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \Phi(t) + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \phi^T(t) \end{aligned}$$

- Finally,

$$\begin{aligned}\bar{Z}(t) &= \Phi^*(t)\, Z(t) \\ &= \begin{pmatrix} \Phi_{22}(t) & -\Phi_{12}(t) \\ -\Phi_{21}(t) & \Phi_{11}(t) \end{pmatrix} Z(t)\end{aligned}$$

In summary,

$$\begin{aligned}\dot{Z}_1(t) &= -Z_1(t) + z(t) \\ \dot{Z}_2(t) &= -2Z_2(t) + 2z(t) \\ \dot{\Phi}(t) &= \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}\Phi(t) + \begin{pmatrix} 1 \\ 2 \end{pmatrix}\phi^T(t) \\ \bar{Z}(t) &= \begin{pmatrix} \Phi_{22}(t) & -\Phi_{12}(t) \\ -\Phi_{21}(t) & \Phi_{11}(t) \end{pmatrix} Z(t)\end{aligned}$$

To apply the gradient method, set

$$\begin{aligned} e &= \hat{\bar{Z}} - \bar{Z} \\ &= \det\Phi(t)\hat{\theta} - \bar{Z} \\ \dot{\hat{\theta}} &= \begin{pmatrix} -\det\Phi(t)\gamma_1 e_1 \\ -\det\Phi(t)\gamma_2 e_2 \end{pmatrix} \\ &= -\det\Phi(t)\begin{pmatrix} \gamma_1 e_1 \\ \gamma_2 e_2 \end{pmatrix}\end{aligned}$$

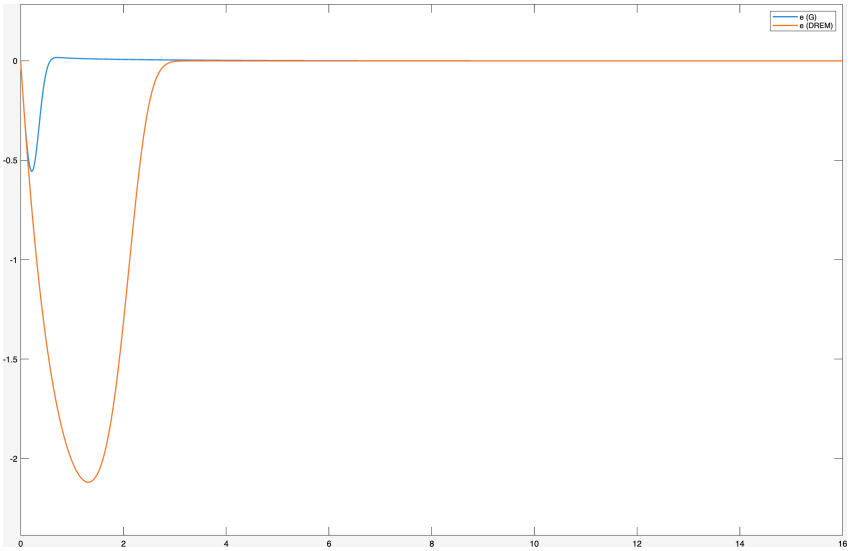
The total error is then

$$e = \hat{\theta}^T\phi - z$$

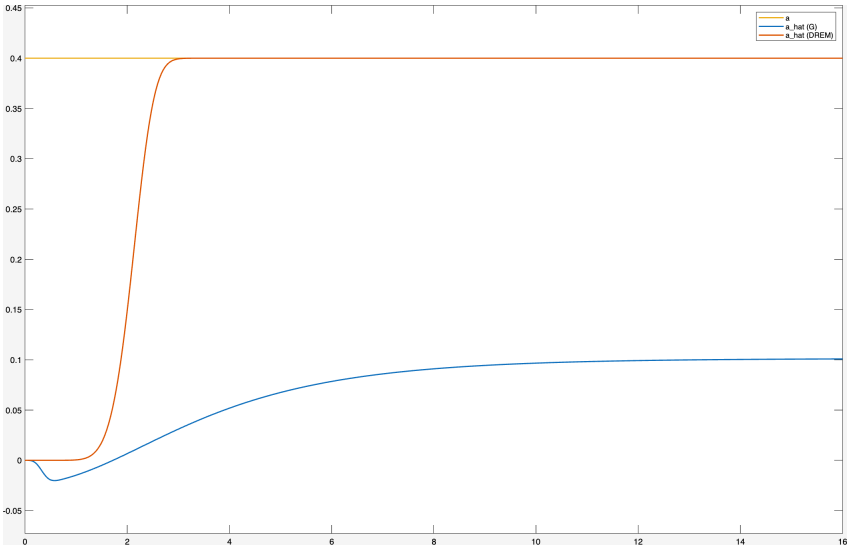
7 Constant input $u = 10$

For the constant input $u = 10$, the results are as follows.

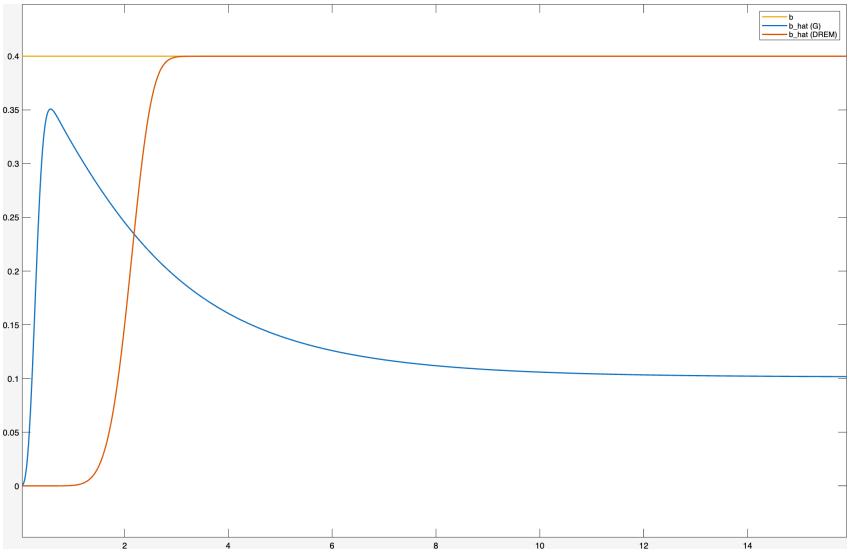
- With the gradient method, the error reaches an initial extremum of -0.56 and converges to zero in about $1s$.
  With DREM, the initial excursion is considerably larger, reaching -2.12, and convergence takes about $3.5s$.



- With the gradient method, the estimate $\hat{a}$ of a converges to $0.1 < a$ in about $13s$, after an undershoot of -0.02.
  With DREM, the estimate $\hat{a}$ of a converges to the true value of a in about $3s$.

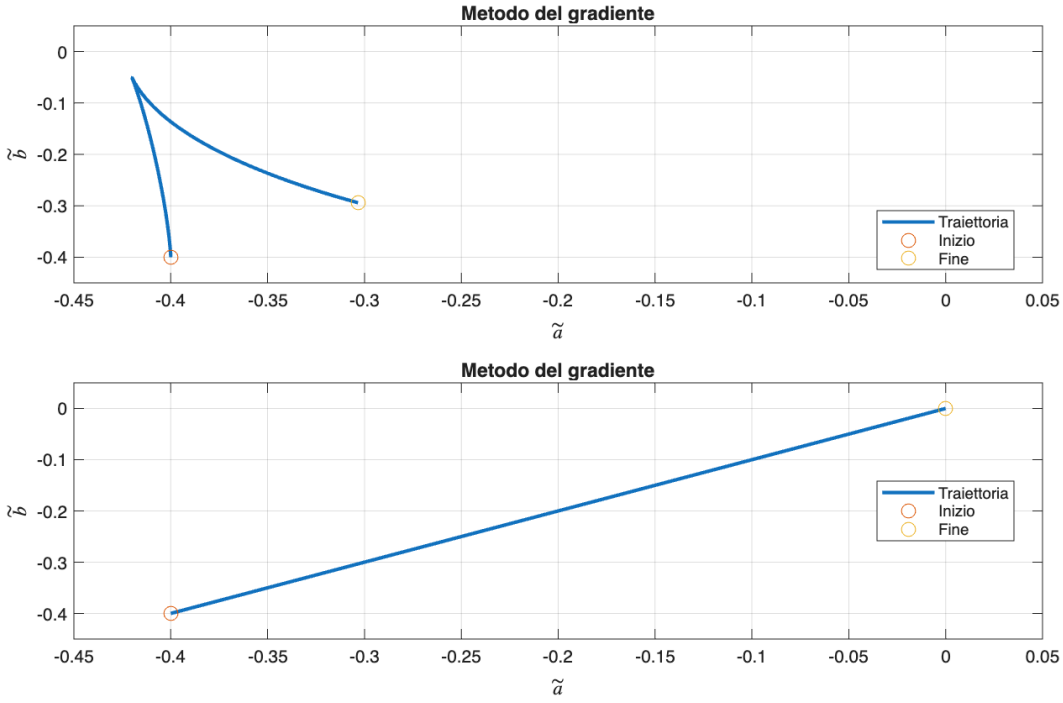


- With the gradient method, the estimate for $b\,\hat{b}$ reaches an initial peak of 0.35 and converges to 0.1 in about $14s$.
  With DREM, the estimate $\hat{b}$ of b shows no initial overshoot and converges to the true value of b in about $3s$.



- With the gradient method, the estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ is curved because the error dynamics are coupled. A change in one parameter affects the evolution of the other. The trajectory converges to $(\tilde{a}(t), \tilde{b}(t)) \approx (-0.3, -0.3)$ because the signal is not persistently exciting.

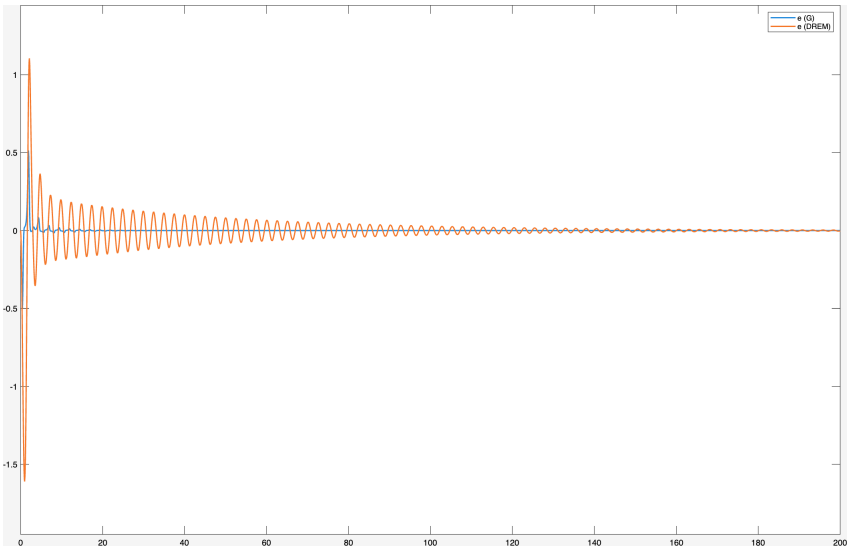
  With DREM, the estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ is approximately straight because the error dynamics are decoupled. The trajectory converges to the origin because $\det \Phi(t) \notin L_2$.



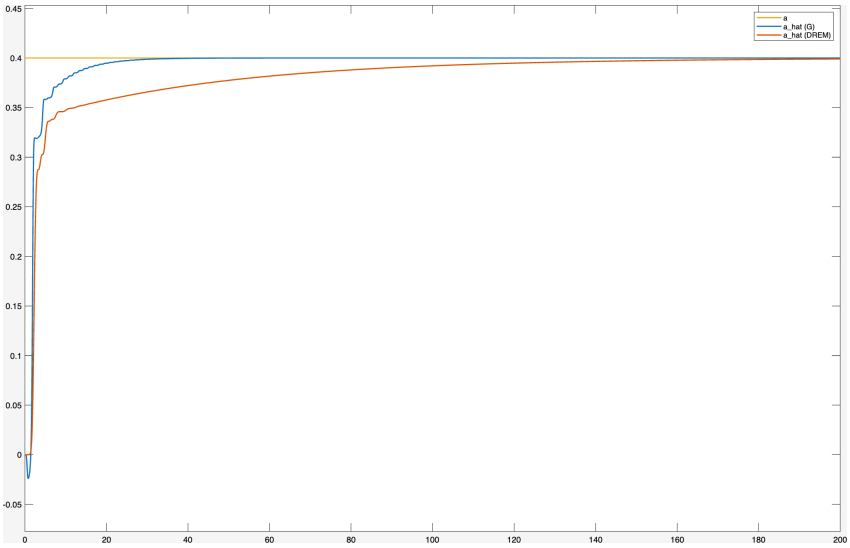
8 Sinusoidal input $u = 10\sin\left(\frac{5t}{2}\right)$

For the sinusoidal input $u = 10\sin(\frac{5t}{2})$, the results are as follows.

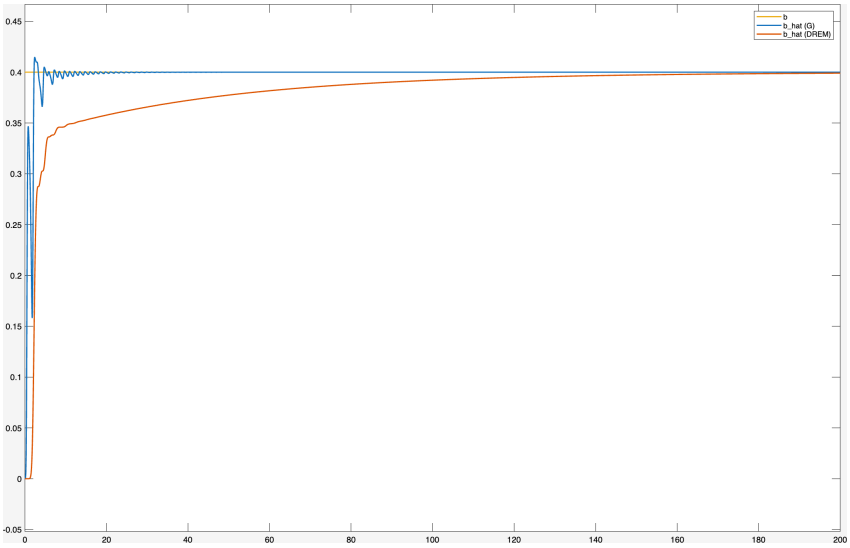
- With the gradient method, the initial error oscillations have a maximum amplitude of 0.5. The error converges to zero in about $30s$.
  With DREM, the initial oscillations have a maximum amplitude of 1.6, and convergence takes about $200s$.



- With the gradient method, the estimate $\hat{a}$ of a converges to the true value of a in about $40s$.
  With DREM, the estimate $\hat{a}$ of a converges to the true value of a in about $200s$.

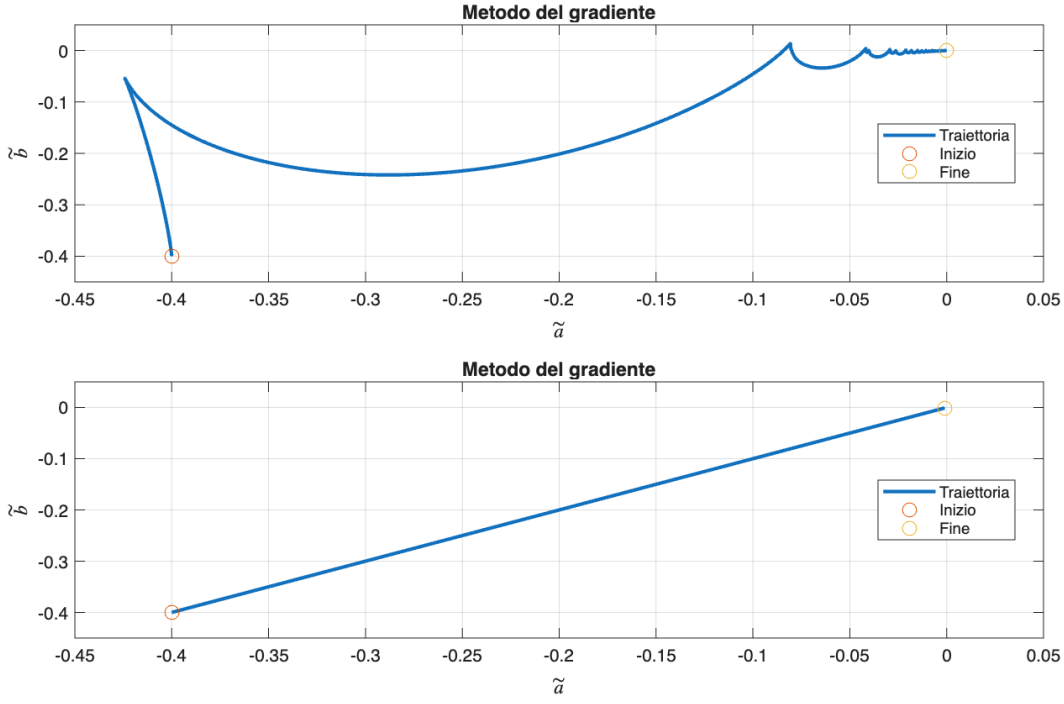


- With the gradient method, the estimate for $b\,\hat{b}$ shows an undershoot of 0.16 and several oscillations before converging to the true value of b in about $40s$.
  With DREM, the estimate $\hat{b}$ of b shows no oscillations and converges to the true value of b in about $200s$.



- With the gradient method, the estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ is curved because the error dynamics are coupled. A change in one parameter affects the evolution of the other. The trajectory converges to the origin because the signal is persistently exciting.

  With DREM, the estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ is approximately straight because the error dynamics are decoupled. The trajectory converges to the origin because $\det \Phi(t) \notin L_2$.



9 Conclusions

The simulations yielded the following observations.

- **In the simulations, the DREM estimation error e shows a larger overshoot than the gradient method.** The filters used to construct the extended regression introduce a delay during the estimator's transient response.

- **DREM converges more slowly than the gradient method.** This behaviour is attributed to the filters $H_1(s)$ and $H_2(s)$ used to construct the extended regression.

- **The gradient estimates of $\hat{a}$ and $\hat{b}$ exhibit overshoot, while the DREM estimates do not.** The parameter estimation errors are coupled in the gradient method and decoupled in DREM.

- **The estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ is curved with the gradient method and nearly straight with DREM.** This difference reflects the coupling of the parameter estimation errors in the gradient method and their decoupling in DREM.

- **With a constant input, the gradient estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ does not converge to the origin**, because the input is not persistently exciting. **The DREM trajectory converges to the origin**, because $\det(\Phi(t)) \notin L_2$.

- **With a sinusoidal input, the estimation error trajectory $(\tilde{a}(t), \tilde{b}(t))$ converges to the origin for both methods.** For the gradient method, the input is persistently exciting. For DREM, convergence follows because $\det(\Phi(t)) \notin L_2$.

- **For both methods, the estimation errors e oscillate more strongly with a sinusoidal input than with a constant input.** The regressor changes continuously over time, altering the direction and magnitude of the parameter updates.